

Why Now: The Safety Analytics Gap

- AMS tools were built to optimize mobility — most models are calibrated to delay and throughput, not to safety
- Connected-vehicle, telematics, and video trajectory data now make proactive, pre-crash analysis practical
- Agencies must justify ITS safety investments without validated methods or accepted guidance



Not "another framework," but a portfolio of safety-focused AMS artifacts that FHWA can post on the forum to help practitioners make measurably better safety decisions

Strategy: Three Principles

▪ 1. Credible Methods

- Standardize surrogate safety measures and safety-specific calibration
- Publish minimum expectations for defensible analysis
- Tie every method to an observable, verifiable safety outcome

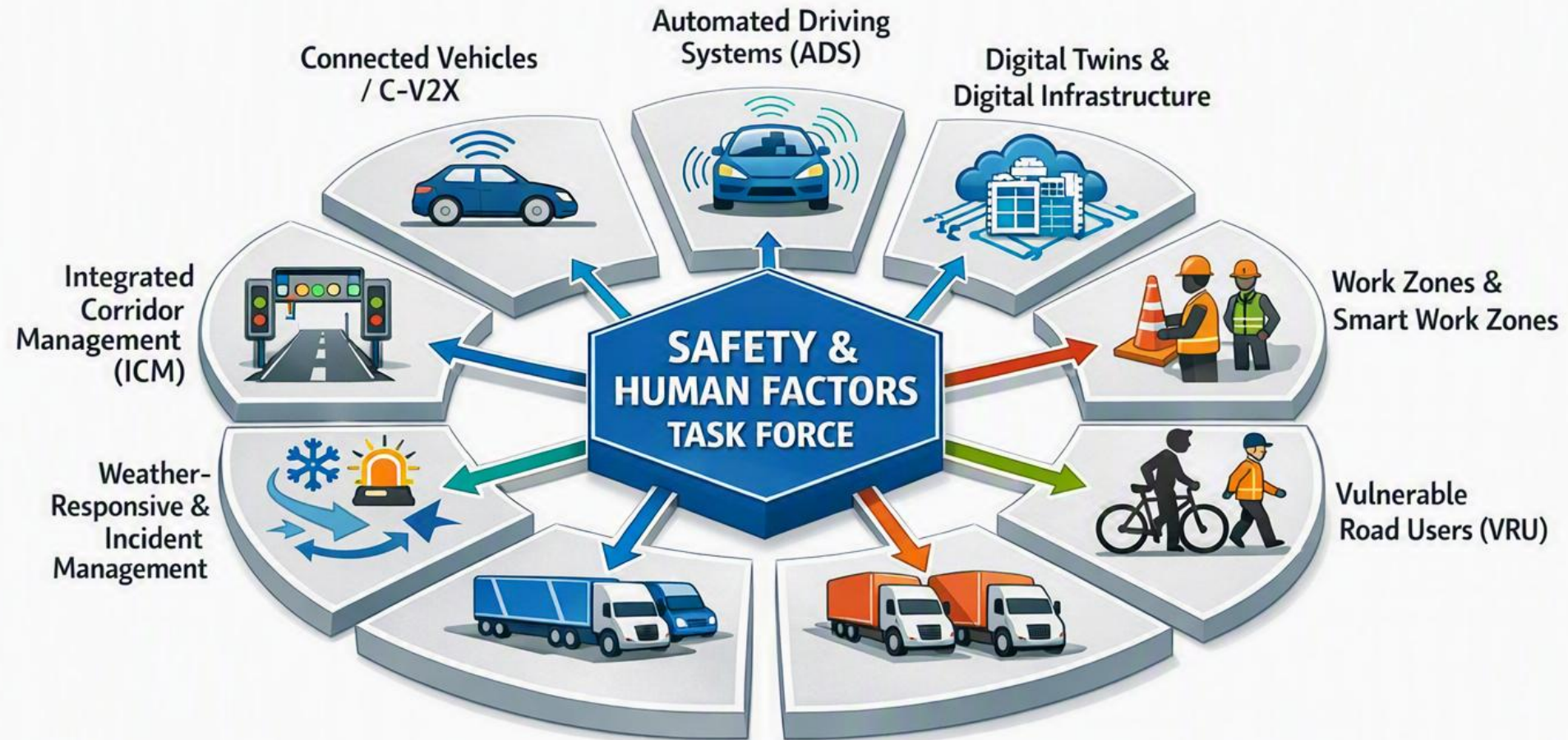
▪ 2. Usable Data

- Curate quality-checked trajectory and connected-vehicle datasets
- Set privacy, sufficiency, and reference-scenario ground rules
- Publish common test corridors so results can be compared

▪ 3. Practitioner Uptake

- Deliver open tools, pilots, and training with every product
- Measure success by agency adoption, not publications
- Pair each release with training and a worked example

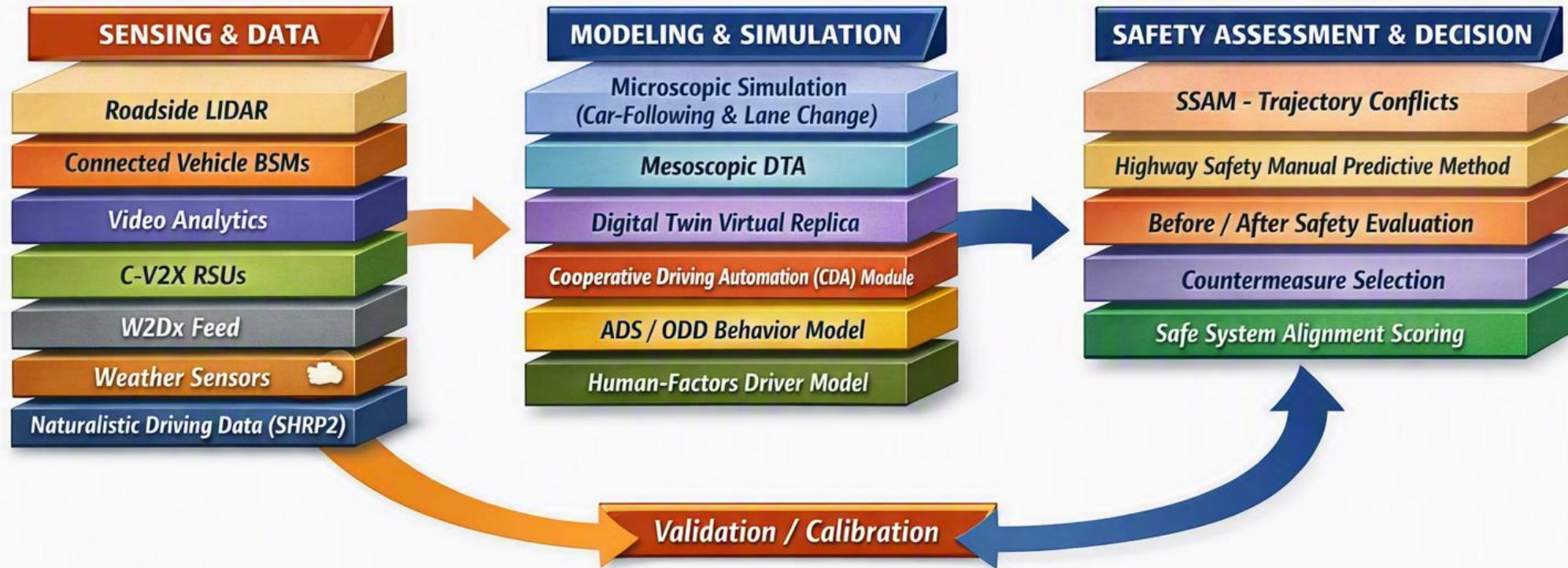
ITS AMS Safety Task Force — Near-Term Portfolio Taxonomy



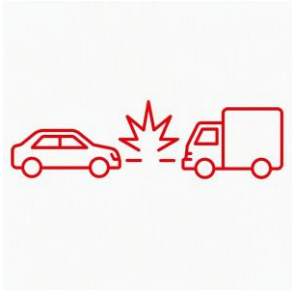
Cross-Cutting Foundations:

Surrogate Safety Measures • Calibration & Validation • Human Factors • Multimodal Safety • Datasets & Standards
Practitioner Guidance

AMS-Enabled Safe System — From Sensor to Countermeasure



Priority Topics: A Balanced Near-Term Portfolio



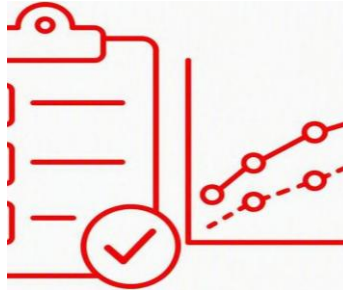
Surrogate Safety Measures

Standardize conflict metrics and link them to observed crash outcomes.



Trajectory and CV Data

Curate quality-checked datasets; define sufficiency and privacy ground rules.



Calibration and Validation

Safety-specific targets, acceptance criteria, and honest uncertainty reporting.



Automated and Mixed Fleets

Behavior models and test scenarios for AVs sharing roads with human drivers.



Vulnerable Road Users

Pedestrian and bicyclist modeling at crossings, intersections, and work zones.



Workforce and Guidance

Training, reference workflows, and peer exchange to build agency capacity.